CLOVERS AKITOYE

Pharmacist (BPharm) | MSc AI for Healthcare | Clinical ML, medical imaging and language models
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PROFILE

Pharmacist completing an MSc in AI for Healthcare at the University of Hull and building AI tools in Pharma Technical Quality at Roche. MSc work spans a vision-language model for 3D chest CT, clinical machine learning on tumour mutation and diagnostic data, and medical image processing, with 85% in AI as a Medical Device. Tests every model the way a pharmacist checks a prescription before it reaches a patient: with controls, traceable numbers and a focus on the error that causes harm. Clinical grounding from hospital pharmacy, a national medical store and a pharmaceutical inspectorate.

EXPERIENCE

PTQ Intern, AI Agent Development | Roche Products Ltd, Pharma Technical Quality **Jul 2026 – Dec 2026**
Opening Up Roche summer internship, extended full time to December 2026. Welwyn Garden City, UK (hybrid).

- Building an AI assistant that helps drug development project teams see which quality and regulatory requirements apply to them, and when, so they are not discovered late.
- Working with quality and operations subject matter experts to fill gaps and check the output against how the work is done, with a qualified person making every GxP decision.
- Testing the assistant with its users before handover. Internship extended full time to take the work further.

Intern Pharmacist | Federal Capital Territory Administration (FCTA), Abuja, Nigeria **Jun 2024 – Jun 2025**

- Rotated through FCT general and district hospitals: dispensing, medication reconciliation and adverse drug reaction monitoring in family medicine, paediatrics, HIV (antiretroviral therapy) and obstetrics and gynaecology clinics.
- Abuja Central Medical Stores: tracked stock across six warehouses that still ran on handwritten tally cards, and flagged discrepancies in distribution records.
- Pharmaceutical inspectorate: supported premises inspections against NAFDAC and Pharmacy Council of Nigeria standards and kept documentation audit-ready.

PROJECTS

Structured radiology report generation from 3D chest CT | MSc dissertation, 2026
github.com/maryclovers/chest-ct-report-generation

- Built a vision-language pipeline on 1,341 RadGenome-ChestCT volumes: frozen ResNet-34, VGG-16 or Med3D encoders, a Q-Former-style cross-attention bridge and BioGPT-Large adapted with low-rank adaptation (LoRA), trained on an NVIDIA A100 on the University of Hull HPC cluster.
- Tested whether the model used the image, with blank-input and shuffled-scan controls, three seeds and a paired bootstrap over 409 test studies. Every encoder scored 0.09 to 0.12 BLEU-4, and so did the blank-input model, which placed the bottleneck in the frozen encoder and set the next step: fine-tuning or a CT-native encoder.

Glioma grade classification from tumour mutations | AI as a Medical Device (module mark 85%), 2026
github.com/maryclovers/tcga-glioma-grading

- Leak-free scikit-learn pipeline on 861 TCGA patients: feature selection inside cross-validation reduced 20 mutation markers to a 10-gene panel. Logistic regression caught 64 of 73 glioblastomas in the held-out test set (sensitivity 0.88, specificity 0.83, AUC 0.90).
- Chose the model on missed-cancer risk, not accuracy, reported the threshold trade-off for the clinical team, and wrote the evidence generation strategy: MHRA software as a medical device pathway, bias and equity, NHS implementation, health economics and post-market surveillance.

Breast cancer classification workflow | AI as a Medical Device (module mark 85%), 2026
github.com/maryclovers/wbc-breast-cancer-classification

- Imbalance handling, four feature selection methods, Naive Bayes, random forest, SVM and a PyTorch network on the Wisconsin dataset. Tuned SVM reached 0.94 cross-validated accuracy, and a 0.3 decision threshold caught every malignant case in the held-out split at 0.84 specificity.

Brain MRI image processing case study | Medical Imaging and AI, 2026
github.com/maryclovers/medical-image-processing

- Classical pipeline on a T1 brain MRI with a focal lesion, every step measured with SNR, CNR, PSNR and SSIM: bilateral denoising cut white-matter noise by 41%, and CLAHE preserved structure where histogram equalisation inverted the

lesion contrast (SSIM 0.77 against 0.48). Backed by nine lab sessions covering DICOM, Hounsfield units, CT reconstruction, MRI and microscopy.

Also on GitHub: a vehicle damage CNN with class weighting (macro recall 0.69 on 1,440 test images), customer churn and segmentation models, and a full-stack invitation web app (Next.js, TypeScript, Supabase).

SKILLS

Machine learning: Python, scikit-learn (pipelines, cross-validation, GridSearchCV, permutation importance), PyTorch, TensorFlow/Keras, Hugging Face Transformers, PEFT/LoRA, MONAI

Medical imaging: DICOM, pydicom, nibabel, OpenCV, scikit-image, Hounsfield units, CT reconstruction, MRI preprocessing, ImageJ, OHIF Viewer

Language models: retrieval-based extraction, prompt pipelines with human review, BioGPT, LoRA fine-tuning

Evaluation: sensitivity and specificity, ROC/AUC, threshold analysis, BLEU, ROUGE, METEOR, BERTScore, control arms, bootstrap intervals

Tools: Git and GitHub, Jupyter, Google Colab, Slurm on HPC (NVIDIA A100), Next.js, TypeScript, Supabase, Google Workspace

Healthcare and regulation: GxP, MHRA software as a medical device pathway, UK GDPR, pharmacovigilance, medication reconciliation, adverse drug reaction reporting

EDUCATION

MSc Artificial Intelligence for Healthcare | University of Hull **2025 – 2026 (award expected Jan 2027)**

Modules: AI as a Medical Device (85%); Fundamentals of Data Science (Distinction); Medical Imaging and AI; Research and Application in AI and Data Science; Understanding AI; Introduction to Programming

Dissertation: Structured Radiology Report Summarisation from Chest CT Using RadGenome-ChestCT

Bachelor of Pharmacy (BPharm) | Bingham University, Nigeria **2017 – 2023**

Faculty representative; dean-selected MC for faculty events; public health outreach with the PANS public health team.

ADDITIONAL INFORMATION

Certifications: Basic Life Support; ACT counter-terrorism awareness; Roche compliance training in AI governance, safety, health and environment (SHE) and records management

Languages: English (fluent), Yoruba (native)

Part-time work: Customer Assistant, Sainsbury's, Hull (Oct 2025 – present)